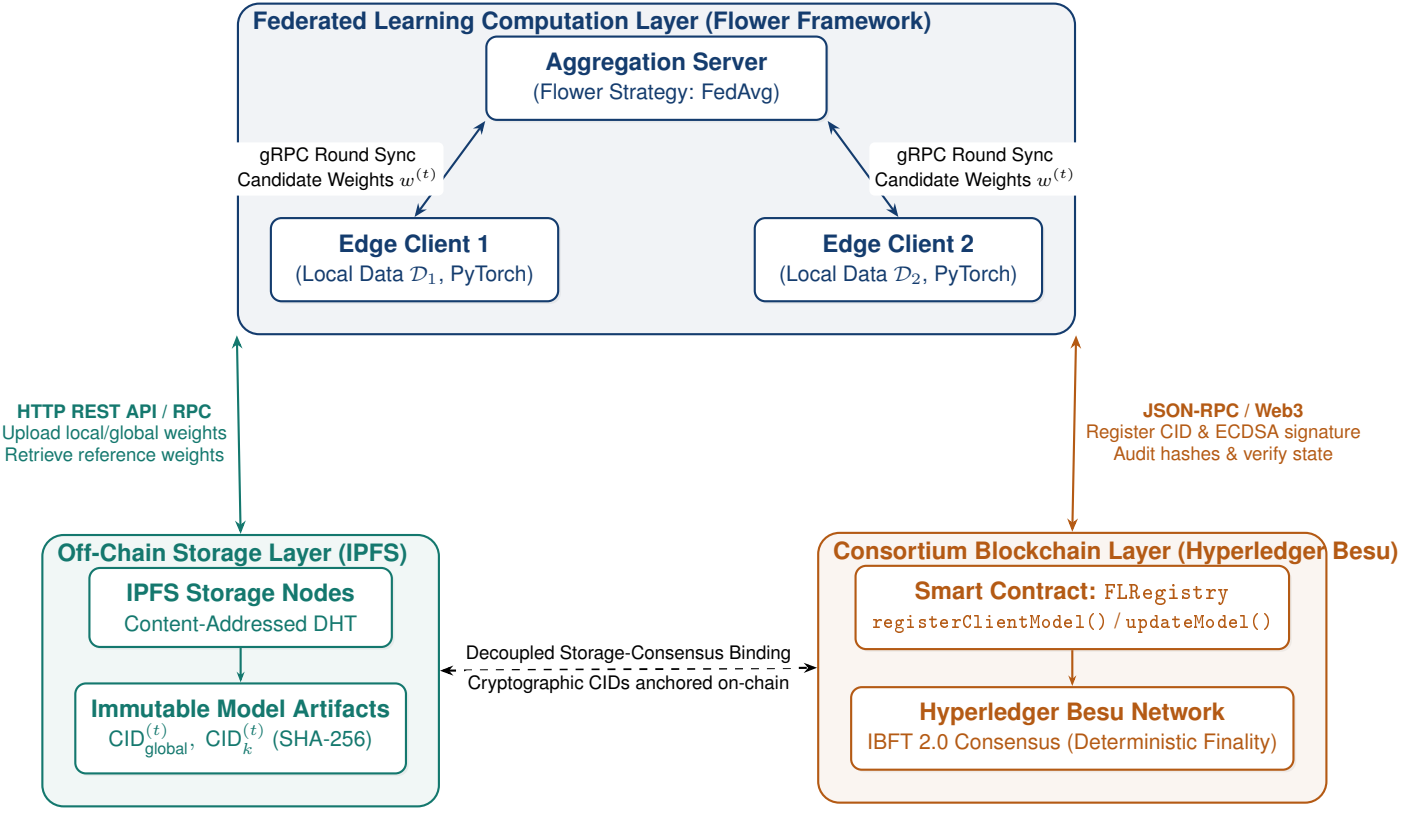

Federated Learning Computation Layer (Flower Framework)
Aggregation Server
(Flower Strategy: FedAvg)
gRPC Round Sync
Candidate Weights $w^{(t)}$
gRPC Round Sync
Candidate Weights $w^{(t)}$
Edge Client 1
(Local Data $\mathcal{D}_1$, PyTorch)
Edge Client 2
(Local Data $\mathcal{D}_2$, PyTorch)
HTTP REST API / RPC
Upload local/global weights
Retrieve reference weights
JSON-RPC / Web3
Register CID & ECDSA signature
Audit hashes & verify state
Off-Chain Storage Layer (IPFS)
IPFS Storage Nodes
Content-Addressed DHT
Immutable Model Artifacts
$\text{CID}_{\text{global}}^{(t)}$, $\text{CID}_{k}^{(t)}$ (SHA-256)
Decoupled Storage-Consensus Binding
Cryptographic CIDs anchored on-chain
Consortium Blockchain Layer (Hyperledger Besu)
Smart Contract: FLRegistry
registerClientModel() / updateModel()
Hyperledger Besu Network
IBFT 2.0 Consensus (Deterministic Finality)